

Generative AI-Driven Knowledge Base Question Answering with Differentiable Knowledge Graphs for Sustainable Engineering Education

1st Sami Ahmed Haider

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
S.Haider@hw.ac.uk

2nd Mukesh Soni

Division of Research and Development
Lovely Professional University
Phagwara, India
Mukesh.soni@gmail.com

3rd Jehan Akbar

School of Engineering
University of the West of England
Bristol, United Kingdom
Jehan.akbar@uwe.ac.uk

4th Samar Abbas

School of Information and Control Engineering
Southwest University of Science and Technology
Mianyang, P.R. China
sammarhashmi@gmail.com

5th Khwaja Muthair Ahmad

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
muthair@swjtu.edu.cn

6th Faisal Tariq

James Watt School of Engineering
University of Glasgow
Glasgow, United Kingdom
Faisal.Tariq@glasgow.ac.uk

7th Masood-ur-Rehman

James Watt School of Engineering
University of Glasgow
Glasgow, United Kingdom
masood.urrehman@glasgow.ac.uk

8th Adnan Zahid

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
a.zahid@hw.ac.uk

Abstract—Knowledge Base Question Answering (KBQA) is a challenging and widely studied research direction. Approaches that rely on embedding use implicit reasoning to obtain answers, but they are not well suited to producing complete, transparent reasoning paths. Models built upon differentiable knowledge graphs, by contrast, can deliver explainable outputs with minimal supervision signals, requiring only question–answer pairs. Building on differentiable knowledge graphs, we propose an encoder–decoder model called EDKBQA. The encoder fine-grainedly and sequentially models questions using a multi-head attention mechanism and an LSTM to generate query vectors that better capture the semantic properties of each reasoning hop. The decoder incorporates a feedforward neural network with an attention mechanism for multi-hop reasoning, effectively representing the relative importance of each hop when determining the final answer. The proposed model directly addresses information loss arising from coarse-grained, non-sequential modelling. Experiments on the MetaQA-1hop, MetaQA-2hop, MetaQA-3hop, WebQSP, and CWQ datasets achieve accuracies of 97.5%, 100%, 100%, 77.8%, and 51.4%, respectively. Ablation studies confirm that every module contributes to overall performance. Furthermore, the paper demonstrates the model’s practical utility through a case study in sustainable engineering education, showcasing its ability to provide transparent, multi-hop explanations for complex engineering concepts.

Index Terms—Generative AI, Multi-Hop Knowledge, Question Answering, Differentiable Knowledge Graph, LSTM, Encoder–Decoder.

I. INTRODUCTION

Knowledge Base Question Answering (KBQA) [1] has become an important technology widely used in generative AI-based applications such as search engines and voice assistants. Improving the accuracy and interpretability of KBQA has therefore become an active research area. A knowledge base (KB) is a data structure that stores facts as triples, using edges to represent relationships between entity nodes; each triple consists of a head entity, a tail entity, and the relationship connecting them. By searching the KB, a KBQA system retrieves the answer to a natural-language query. The two principal categories are single-hop KBQA, in which the subject and answer are connected by a single head–relation–tail triple, and multi-hop KBQA, in which the subject may be several hops away from the answer.

In single-hop KBQA, the straightforward triple structure yields near-human performance. Multi-hop KBQA, however, requires navigating multiple linked triples, substantially increasing semantic complexity and demanding deeper reasoning. As a result, multi-hop KBQA has attracted growing research interest, with current models seeking to improve both accuracy, by identifying the correct answer entity, and interpretability, by explicitly modelling the reasoning path between the subject and the final answer.

Answering complex multi-hop queries requires specialised methods [2]. Existing approaches fall broadly into two cat-

egories: semantic parsing and information retrieval. Semantic parsing methods [3] translate questions into intermediate structured forms—such as query graphs or logical statements—which are then executed over the KB. However, they require costly question-logic annotation and domain expertise in query languages; moreover, the intermediate structured queries do not always faithfully capture the question’s semantics, leading to answer failures. Information retrieval-based methods [4] directly retrieve answers by learning semantic representations of both the question and the KB. These are further divided into embedding-based methods [5] and path-based methods [6].

Embedding-based methods score the embeddings of the subject and candidate entities, but their reasoning is implicit and lacks interpretability. Path-based methods [7] trace explicit reasoning paths from the subject entity to the answer, offering strong interpretability, though they struggle to scale when the KB or path space is large. Multi-hop questions exhibit distinct semantics at each hop and a sequential reasoning path that is both transparent and visualisable [8], with different hops carrying different importance weights when determining the final answer.

This paper presents an end-to-end EDKBQA model built on a differentiable knowledge graph [9], [10]. The encoder employs fine-grained sequential modelling, combining multi-head attention [11] to capture detailed semantics with an LSTM [12] to learn the logical ordering of hops. The decoder employs a multi-hop attention module with two feedforward networks to assign hop-importance weights, thereby enabling accurate semantic retention and effective reasoning. Experiments demonstrate that EDKBQA significantly improves accuracy and interpretability over baseline models, with its fully differentiable architecture ensuring high computational efficiency, strong scalability, and straightforward portability.

The contributions of this paper are summarised as follows:

- We propose an encoder–decoder model based on a differentiable knowledge graph that achieves excellent question-answering performance across three datasets, highly competitive with current state-of-the-art models.
- We develop a fine-grained sequential module that uses an LSTM to propagate information across hops and multi-head attention to capture precise hop-level semantics, producing more coherent and expressive query vectors.
- We design a multi-hop reasoning attention module in the decoder that uses a GRU-enhanced relation scorer and dual feedforward networks to determine hop-importance weights and generate final entity scores.
- We strengthen the link to sustainable engineering education with a concrete case study, demonstrating how the model’s explainable reasoning paths can serve as an AI tutor for complex engineering systems.

II. PROPOSED EDKBQA MODEL

This section first introduces relevant background and task definitions, including knowledge bases, KBQA, and differentiable knowledge graphs, before presenting the encoder–decoder architecture.

A. Background and Definitions

A KB represents factual information as a set of triples:

$$\mathcal{KB} = \{\langle H, R, T \rangle \mid H, T \in \mathcal{E}, r \in \mathcal{R}\},$$

where $\mathcal{E}$ is the entity set and $\mathcal{R}$ is the relation set. A triple $\langle H, R, T \rangle$ indicates that relation R holds between head entity H and tail entity T . The neighbourhood of an entity in the KB is defined as $\mathcal{N} = \{\langle H, R, T \rangle \in \mathcal{KB}\}$, combining all triples—including both incoming and outgoing edges—around each entity. For brevity, an inverse triple $\langle T, R, H \rangle$ is written as $\langle H, R^{-1}, T \rangle$.

A KBQA task [26] requires, given a natural-language question $q = w_1, w_2, \dots, w_l$ and a KB, finding the answer entity set $\mathcal{A}_q$ from the candidate entity set, starting from the subject entities mentioned in q . Multi-hop KBQA handles complex questions where the subject and answer are separated by multiple hops, requiring multi-step reasoning across several entities and relations.

A differentiable knowledge graph [28] represents a KB as a sparse matrix structure, allowing question–KB interaction through differentiable mathematical operations. Formally, $\mathcal{KB} = \{\langle H, P, T \rangle \mid H \in \mathcal{E}, T \in \mathcal{E}, P \in \mathcal{R}\}$, where triples are encoded by three binary matrices: the head-entity matrix $\mathbf{H}_M \in \{0, 1\}^{N_T \times N_E}$, the relation matrix $\mathbf{P}_M \in \{0, 1\}^{N_T \times N_R}$, and the tail-entity matrix $\mathbf{T}_M \in \{0, 1\}^{N_T \times N_E}$, defined as:

$$\mathbf{H}_M(i, j) = \mathbb{I}[e_j = s_i], \quad \mathbf{P}_M(i, j) = \mathbb{I}[p_j = p_i], \quad \mathbf{T}_M(i, j) = \mathbb{I}[e_i = o_i]$$

Given an entity score vector $\mathbf{a}_{t-1} \in \mathbb{R}^{N_E}$ and a relation score vector $\mathbf{r}_t \in \mathbb{R}^{N_R}$ at hop $t - 1$, the entity score vector at hop t is computed as:

$$\mathbf{a}_t = f(\mathbf{a}_{t-1}, \mathbf{r}_t). \quad (1)$$

An alternative representation [26] uses a fact matrix $\mathbf{Z} \in \mathbb{R}^{N_E \times N_E}$, where element $Z_{i,j}$ encodes the relationship between head entity E_i and tail entity E_j . Specifically:

$$Z_{i,j}^{(t)} = \sum_b \text{pr}_{i,j,b}^{(t)}, \quad (2)$$

where $\text{pr}_{i,j,b}^{(t)}$ is the b -th probability value of the relation between E_i and E_j at hop t . The entity score vector at hop t is then:

$$\mathbf{a}_t = \mathbf{Z}^{(t)\top} \mathbf{a}_{t-1}. \quad (3)$$

B. Model Architecture

The EDKBQA model leverages a differentiable knowledge graph to enable fully end-to-end reasoning, executing multi-hop question–KB interactions through differentiable, generative AI operations. This path-based design is both computationally efficient and capable of explicitly modelling intermediate reasoning paths. The model can also be parallelised across multiple GPUs, providing strong interpretability and excellent scalability.

Fig. 1 illustrates the EDKBQA architecture, which comprises an encoder for question representation and a decoder for

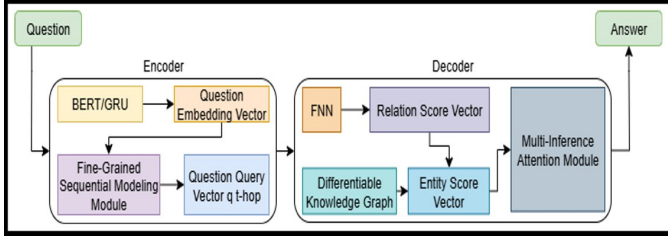


Fig. 1. Encoder-Decoder Model Architecture.

multi-hop reasoning. The encoder produces a per-hop query vector through a fine-grained sequential module that combines multi-head attention for detailed semantic representation and an LSTM for capturing hop-level logical ordering. The decoder then reasons over these query vectors to derive the final answer.

The model first derives a per-hop relation score vector from the query representation, then computes a per-hop entity score vector through differentiable knowledge graph operations, and finally aggregates these to produce the final entity score used to select the answer. The multi-hop attention module employs two feedforward networks to weight hop importance.

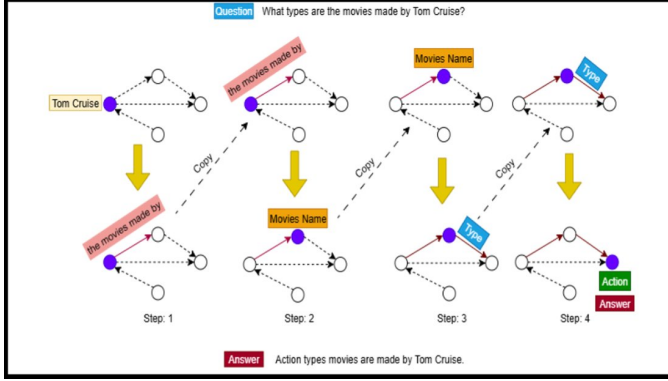


Fig. 2. Multi-Hop Reasoning Process.

Fig. 2 illustrates the multi-hop reasoning process using the example question “What types of movies are made by Tom Cruise?” The model uses named entity recognition (NER) to identify the subject entity “Tom Cruise” and links it to the corresponding KB entity. It then initialises relation and entity score vectors, and iteratively refines these over multiple reasoning hops. In the final step, the decoder’s multi-hop attention module computes importance weights for each hop and takes a weighted sum of the per-hop entity score vectors to identify the highest-scoring answer entity, “Action”.

For questions requiring more than two hops, the single-hop reasoning step is simply repeated until the answer is reached, giving the model clear interpretability through explicit intermediate paths. Its fully differentiable knowledge graph further ensures strong scalability and efficiency.

C. Encoder

The encoder’s task is to generate a per-hop query vector that better represents the semantic information at each reasoning step. Existing models tend to produce coarse-grained, sentence-level representations that lose key hop-level information and ignore the sequential logical dependencies between hops. To address this, we introduce an encoder that produces fine-grained, hop-aware query vectors by integrating past and current reasoning states via an LSTM, while multi-head attention captures precise semantic details at the token level.

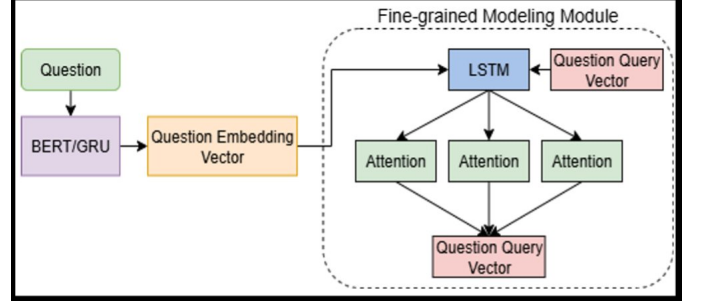


Fig. 3. Encoder Architecture.

Fig. 3 shows the encoder architecture. First, a pre-trained language model (BERT or GRU) generates the initial question embedding $\mathbf{k}$ and token-level embeddings $\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_n$:

$$\mathbf{k}, \mathbf{t}_1, \dots, \mathbf{t}_n = \text{Encoder}(q; \theta_e). \quad (4)$$

The LSTM then takes the question embedding $\mathbf{k}$ and the query vector from hop $t-1$ to generate the query vector for hop t :

$$\mathbf{k}_t = \text{LSTM}(\mathbf{k} \mid \mathbf{k}_{t-1}; \theta_w). \quad (5)$$

The query vector $\mathbf{k}_t$ serves as the query Q matrix, while the token embeddings serve as both the key K and value V matrices. Multi-head attention computes:

$$b_j^{(t)} = \text{Softmax}(\mathbf{k}_t \cdot [\mathbf{t}_1, \dots, \mathbf{t}_n]^\top), \quad (6)$$

$$\hat{\mathbf{k}}_t^{(i)} = \sum_{j=1}^n b_j^{(t)} \mathbf{t}_j. \quad (7)$$

Finally, the outputs of m attention heads are averaged to obtain the hop-level query vector:

$$\mathbf{k}_t = \frac{1}{m} \sum_{i=1}^m \hat{\mathbf{k}}_t^{(i)}. \quad (8)$$

D. Decoder

The decoder transforms the encoder’s query vectors into candidate entity scores through differentiable operations on the knowledge graph. For each hop, it generates a relation vector from the query vector, then combines it with the previous hop’s entity vector via differentiable knowledge graph reasoning to produce the next hop’s entity vector. A weighted sum of all hop-level entity vectors forms the final score vector.

The multi-hop reasoning attention module uses two feed-forward neural networks to learn the relative importance of different hops. One network is trained on the hop-specific query vectors; the other is trained on the global question embedding. The final hop-importance weight is obtained by averaging the two independently computed weights, combining local hop semantics with global question semantics to yield more reliable entity score estimates.

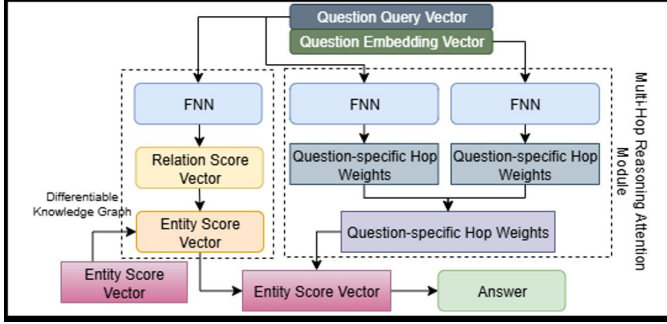


Fig. 4. Decoder Architecture.

Fig. 4 illustrates the decoder. The operations are summarised as follows. For each hop t , a feedforward network transforms the query vector into a relation score vector:

$$\mathbf{r}_t = \text{FNN}(\mathbf{k}_t). \quad (9)$$

Differentiable knowledge graph operations then produce the hop-level entity score vector from the previous entity vector $\mathbf{w}_{t-1}$ and $\mathbf{r}_t$ (only the topic entity is assigned a score of 1 in $\mathbf{w}_0$):

$$\mathbf{w}_t = f(\mathbf{w}_{t-1}, \mathbf{r}_t). \quad (10)$$

Two feedforward networks compute hop-importance weights from the global embedding $\mathbf{k}$ and the per-hop query $\mathbf{k}_t$, respectively:

$$\tilde{w}_t^{(2)} = \text{FNN}(\mathbf{k}), \quad (11)$$

$$\tilde{w}_t^{(1)} = \text{Softmax}(\text{FNN}(\mathbf{k}_t)). \quad (12)$$

These are averaged to obtain the final hop weight:

$$\tilde{w}_t = \frac{\tilde{w}_t^{(1)} + \tilde{w}_t^{(2)}}{2}. \quad (13)$$

The final entity score vector is then the weighted sum across all hops:

$$\mathbf{ev} = \sum_{t=1}^T \mathbf{w}_t \tilde{w}_t. \quad (14)$$

The entity with the highest score in $\mathbf{ev}$ is selected as the answer.

III. EXPERIMENTS

A. Datasets

We evaluate our approach on MetaQA, WebQuestionsSP (WebQSP), and ComplexWebQuestions (CWQ), with statistics shown in Table I. MetaQA is a large multi-hop KGQA

benchmark derived from WikiMovies, containing nearly 400k questions (up to 3 hops) built over a movie-domain knowledge graph with 43k entities, 9 predicates, and 135k triples. WebQSP is a refined subset of WebQuestions providing 4,737 questions linked to Freebase; unanswered questions are removed during training and counted as errors during evaluation. All WebQSP questions can be answered via one- or two-hop reasoning. Following prior work, we prune Freebase to a subgraph containing all facts within two hops of the topic entity, yielding a processed KB with 1.8M entities, 572 predicates, and 5.7M triples.

TABLE I
SUMMARY OF BENCHMARK DATASETS

Dataset	Train	Validation	Test
MetaQA-1hop	97,127	9,996	9,965
MetaQA-2hop	119,335	14,976	14,872
MetaQA-3hop	114,245	14,307	14,274
WebQSP	3,125	100	1,705
CWQ	27,525	3,518	3,631

CWQ extends WebQSP with deeper hops and richer constraints. Following the approach in [31], we pre-process CWQ using PageRank to retrieve a question-specific subgraph for each query [32], yielding subgraphs averaging 1,948 entities with a 64% answer recall rate. Experiments on CWQ are conducted on the validation set.

B. Baselines

Our model is compared against the following baseline methods: KGT5 [16], GraftNet [17], PullNet [18], Relation-Learning [19], EMQL [20], NSM [23], SR+NSM [24], Rigel-Baseline [25], TransferNet [26], CFG [28], and Struct-GPT [30].

C. Evaluation Metrics

Hits@1 ranks all candidate entities by score and assigns 1 if the correct answer is ranked first, and 0 otherwise. As the strongest indicator of a model's ability to identify the correct entity, Hits@1 is averaged over all test questions, with unanswered cases counted as errors. This paper adopts Hits@1 as the primary evaluation metric.

F1 treats all entities scoring above a chosen threshold as predicted answers and computes the harmonic mean of precision and recall against the ground truth. The threshold is tuned on the validation set. Because F1 is highly sensitive to the threshold and many baselines do not report it, F1 comparisons are provided only for WebQSP.

D. Experimental Results

The MetaQA dataset uses a bidirectional GRU as the question encoder, whilst WebQSP and CWQ use bert-base-uncased owing to their greater complexity. Hop counts are set to 3, 2, and 3 for MetaQA, WebQSP, and CWQ, respectively. All experiments were run on an NVIDIA RTX 3080 GPU. The model was trained using RAdam (learning rate = 1×10^{-3}); BERT parameters were fine-tuned with

a smaller learning rate of 3×10^{-5} . Multi-head attention used 3 heads.

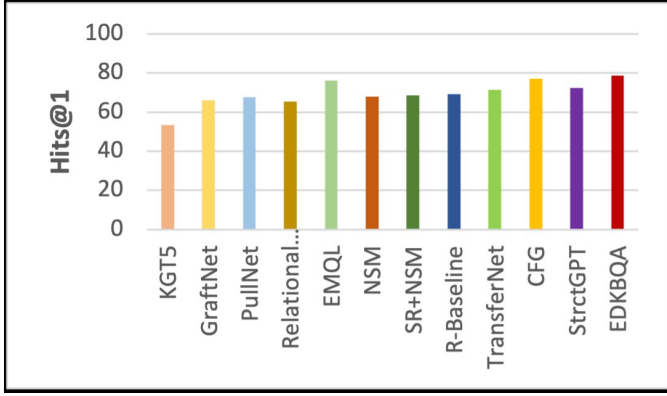


Fig. 5. Hits@1 Results on WebQSP.

Fig. 5 shows the Hits@1 results on WebQSP. On MetaQA, EDKBQA correctly answers all 2-hop and 3-hop questions and achieves 97.5% accuracy on 1-hop queries; minor errors arise from entities sharing identical names. Multi-hop queries benefit from stronger relational constraints. Our model attains 77.8% accuracy on WebQSP and 51.4% on CWQ, both substantially exceeding baseline performance.

The baseline models fall into two categories: embedding-based and path-based methods. Results show that path-based EDKBQA outperforms all embedding-based approaches, demonstrating superior interpretability, scalability, and overall performance. In the encoder, multi-head attention captures fine-grained token semantics whilst the LSTM models hop-to-hop dependencies, producing query vectors that more accurately reflect the question’s meaning. In the decoder, two feedforward networks compute hop-importance weights, allowing the model to capture the contribution of each hop to the final answer.

The EDKBQA model is built on a fully differentiable knowledge graph, making it computationally efficient, fast to train, scalable across multiple GPUs, and easily extensible without retraining when new entities are added. Analysis of failure cases reveals two main issues: (i) the encoder occasionally misidentifies the subject entity, preventing correct relation-path discovery—an issue mitigable by adding missing entities to BERT’s vocabulary; and (ii) semantically similar relations with limited training examples cause the model to confuse relation types and fail to derive the correct answer path.

Recent work has explored how large language models (LLMs) perform on multi-hop KBQA; our experiments show that EDKBQA substantially outperforms StructGPT. LLMs that rely solely on their internal knowledge perform poorly on specialised KB reasoning and, even with fine-tuning, remain less stable and less reliable than dedicated KBQA models.

E. Ablation Study

Fig. 6 shows the ablation results. Removing any component—the encoder’s LSTM, the multi-head attention

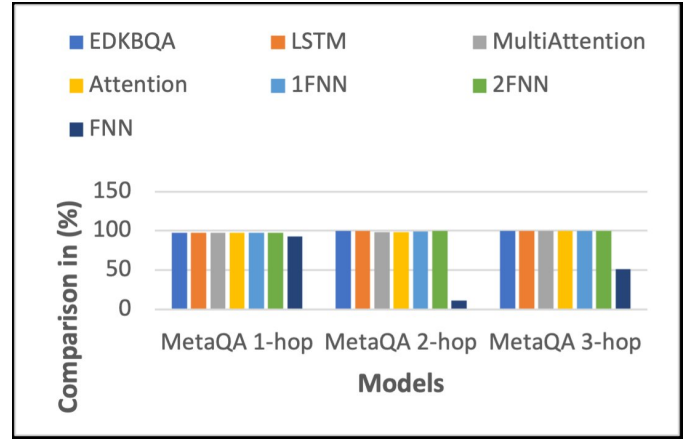


Fig. 6. Ablation Study Results on MetaQA.

mechanism, or either of the decoder’s two feedforward networks—causes substantial performance drops across all datasets. Replacing multi-head attention with a single-head variant also noticeably weakens results, underscoring the value of fine-grained sequential and semantic modelling. The encoder’s LSTM and multi-head attention, together with the decoder’s dual feedforward networks, deliver the largest performance gains on the most challenging multi-hop tasks.

IV. CONCLUSION

This paper proposes EDKBQA, an end-to-end differentiable knowledge graph-based encoder–decoder framework for question answering. The encoder fuses fine-grained sequential modelling with an LSTM to capture hop-wise dependencies, whilst multi-head attention highlights important semantic cues when constructing per-hop query vectors. The decoder improves accuracy with a multi-hop attention module that uses two feedforward networks to estimate hop-importance weights, yielding more reliable entity scores and final answers. Extensive experiments confirm the effectiveness of EDKBQA, demonstrating strong improvements in answer accuracy. Future work will focus on improving relation-vector generation, currently performed by a simple linear classifier in the decoder, to further enhance multi-hop reasoning performance.

REFERENCES

- [1] J. Qadir, “Engineering education in the era of ChatGPT: Promise and pitfalls of generative AI for education,” in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, 2025.
- [2] S. Wang and B. Qin, “A novel joint training model for knowledge base question answering,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 666–679, 2024, doi: 10.1109/TASLP.2023.3336526.
- [3] C. Wang, K. Xue, and C. Shi, “An optimized model with encoder-decoder ConvLSTM for global ionospheric forecasting,” *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025, Art no. 1001605, doi: 10.1109/LGRS.2025.3565645.
- [4] Y. Lin and H. Shen, “SmartQ: A question and answer system for supplying high-quality and trustworthy answers,” *IEEE Trans. Big Data*, vol. 4, no. 4, pp. 600–613, Dec. 2018, doi: 10.1109/TBDATA.2017.2735442.
- [5] A. Serrano-Laguna, I. Martinez-Ortiz, P. Moreno-Ger, and B. Fernandez-Manjon, “AI as yet another tool in undergraduate student projects: Preliminary results,” in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, 2024, pp. 1337–1342, doi: 10.1109/EDUCON60312.2024.10511606.

- [6] U. Naseem, M. Khushi, A. G. Dunn, and J. Kim, "K-PathVQA: Knowledge-aware multimodal representation for pathology visual question answering," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 4, pp. 1886–1895, Apr. 2024, doi: 10.1109/JBHI.2023.3294249.
- [7] Y. Zheng, L. Lu, Y. Hu, Y. Chen, and A. Wang, "Thoughtful and cautious reasoning: A fine-tuned knowledge graph-based multi-hop question answering framework," *Eng. Appl. Artif. Intell.*, vol. 150, p. 110479, 2025, doi: 10.1016/j.engappai.2025.110479.
- [8] Z. Wang, X. Su, and Z. Ding, "Long-term traffic prediction based on LSTM encoder-decoder architecture," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 10, pp. 6561–6571, Oct. 2021, doi: 10.1109/TITS.2020.2995546.
- [9] H. Zhou, T. He, Y.-S. Ong, G. Cong, and Q. Chen, "Differentiable clustering for graph attention," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 3751–3764, Aug. 2024, doi: 10.1109/TKDE.2024.3363703.
- [10] W. Wu, Z. Zhu, J. Qi, W. Wang, G. Zhang, and P. Liu, "A dynamic graph expansion network for multi-hop knowledge base question answering," *Neurocomputing*, vol. 515, pp. 37–47, 2023, doi: 10.1016/j.neucom.2022.09.127.
- [11] J. Ma, J. Liu, Q. Lin, B. Wu, Y. Wang, and Y. You, "Multi-task learning for visual question answering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1380–1394, Mar. 2023, doi: 10.1109/TNNLS.2021.3105284.
- [12] L. Yang, H. Chen, Z. Li, X. Ding, and X. Wu, "Give us the facts: Enhancing large language models with knowledge graphs for fact-aware language modeling," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3091–3110, Jul. 2024, doi: 10.1109/TKDE.2024.3360454.
- [13] H. Ren, Z. Chen, X. Liang, C. Yang, and W. Gui, "Association hierarchical representation learning for plant-wide process monitoring using multilevel knowledge graph," *IEEE Trans. Artif. Intell.*, vol. 4, no. 4, pp. 636–649, Aug. 2023, doi: 10.1109/TAI.2022.3161860.
- [14] Y. Wen, X. Zhu, and L. Zhang, "CQACD: A concept question-answering system for intelligent tutoring using a domain ontology with rich semantics," *IEEE Access*, vol. 10, pp. 67247–67261, 2022, doi: 10.1109/ACCESS.2022.3185400.
- [15] J. Xu, H. Zhang, H. Zhang, J. Lu, and G. Xiao, "ChatTf: A knowledge graph-enhanced intelligent Q&A system for mitigating factuality hallucinations in traditional folklore," *IEEE Access*, vol. 12, pp. 162638–162650, 2024, doi: 10.1109/ACCESS.2024.3485877.
- [16] A. Ait-Mlouk and L. Jiang, "KBot: A knowledge graph based chatbot for natural language understanding over linked data," *IEEE Access*, vol. 8, pp. 149220–149230, 2020, doi: 10.1109/ACCESS.2020.3016142.
- [17] S. Zhang, Z. Hu, A. Subramonian, and Y. Sun, "Motif-driven contrastive learning of graph representations," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 4063–4075, Aug. 2024, doi: 10.1109/TKDE.2024.3364059.
- [18] L. Nie, X. Wei, D. Zhang, X. Wang, Z. Gao, and Y. Yang, "Data-driven answer selection in community QA systems," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 6, pp. 1186–1198, Jun. 2017, doi: 10.1109/TKDE.2017.2669982.
- [19] Z. Li, H. Wang, and W. Zhang, "Translational relation embeddings for multi-hop knowledge base question answering," *J. Web Semant.*, vol. 74, p. 100723, 2022, doi: 10.1016/j.websem.2022.100723.
- [20] H. Xiang, Y. Wang, X. Zhang, and Z. Chen, "Two-level battery health diagnosis using encoder-decoder framework and Gaussian mixture ensemble learning based on relaxation voltage," *IEEE Trans. Transp. Electr.*, vol. 10, no. 2, pp. 3966–3975, Jun. 2024, doi: 10.1109/TTE.2023.3317449.
- [21] S. Hu, L. Zou, J. X. Yu, H. Wang, and D. Zhao, "Answering natural language questions by subgraph matching over knowledge graphs," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 5, pp. 824–837, May 2018, doi: 10.1109/TKDE.2017.2766634.
- [22] H. Liu, T. Wang, C. Lang, Y. Li, and H. Ling, "Deep probabilistic graph matching," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5127–5140, Sept. 2025, doi: 10.1109/TKDE.2025.3578989.
- [23] J. Castro et al., "A big data semantic driven context aware recommendation method for question-answer items," *IEEE Access*, vol. 7, pp. 182664–182678, 2019, doi: 10.1109/ACCESS.2019.2957881.
- [24] H. Li, F. Qi, and G. Shi, "A novel spatio-temporal 3D convolutional encoder-decoder network for dynamic saliency prediction," *IEEE Access*, vol. 9, pp. 36328–36341, 2021, doi: 10.1109/ACCESS.2021.3063372.
- [25] J. Chen, X. Xin, X. Liang, X. He, and J. Liu, "GDSRec: Graph-based decentralized collaborative filtering for social recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 5, pp. 4813–4824, May 2023, doi: 10.1109/TKDE.2022.3153284.
- [26] J. Yu, Q. Su, X. Quan, and J. Yin, "Multi-hop reasoning question generation and its application," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 725–740, Jan. 2023, doi: 10.1109/TKDE.2021.3073227.
- [27] W. Zheng, L. Yan, W. Zhang, L. Ouyang, and D. Wen, "D→K→I: Data-knowledge-driven group intelligence framework for smart service in education metaverse," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 4, pp. 2056–2061, Apr. 2023, doi: 10.1109/TSMC.2022.3228849.
- [28] Y. Feng, Y. Sun, G. G. Yen, and K. C. Tan, "REP: An interpretable robustness enhanced plugin for differentiable neural architecture search," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 5, pp. 2888–2902, May 2025, doi: 10.1109/TKDE.2025.3543503.
- [29] R.-Z. Wang, Z.-H. Ling, and Y. Hu, "Knowledge base question answering with attentive pooling for question representation," *IEEE Access*, vol. 7, pp. 46773–46784, 2019, doi: 10.1109/ACCESS.2019.2909826.
- [30] J. Van der Putten, F. Van der Sommen, and P. H. N. De With, "Efficient decoder reduction for a variety of encoder-decoder problems," *IEEE Access*, vol. 8, pp. 169444–169455, 2020, doi: 10.1109/ACCESS.2020.3020360.
- [31] H. M. Yao, L. Jiang, and M. Ng, "Implementing the fast full-wave electromagnetic forward solver using the deep convolutional encoder-decoder architecture," *IEEE Trans. Antennas Propag.*, vol. 71, no. 1, pp. 1152–1157, Jan. 2023, doi: 10.1109/TAP.2022.3216920.
- [32] R. Li, D. Li, H. Yang, X. Liu, H. Jin, R. Pu, and Q. Miao, "RECoT: Relation-enhanced chains-of-thoughts for knowledge-intensive multi-hop questions answering," *Neurocomputing*, vol. 637, p. 129903, 2025, doi: 10.1016/j.neucom.2025.129903.



Haider, Sami Ahmed, Soni, Mukesh, Akbar, Jehan, Abbas, Samar, Ahmad, Khwaja Muthair, Tariq, Faisal, Ur Rehman, Masood and Zahid, Adnan (2026) Generative AI-Driven Knowledge Base Question Answering with Differentiable Knowledge Graphs for Sustainable Engineering Education. In: 2026 IEEE Global Engineering Education Conference (EDUCON), Cairo, Egypt, 27-30 April 2026, ISBN 9798331576714 (doi: [10.1109/educon67543.2026.11574212](https://doi.org/10.1109/educon67543.2026.11574212))



Copyright © 2026 IEEE

This is the author accepted manuscript reproduced under a [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/).

<https://eprints.gla.ac.uk/390225/>

Deposited on: 23 July 2026